二.1.c) $\Pi_{SName,SA}(\sigma_{PS='小说'}(P) \bowtie SP \bowtie S)$

(2) $\Pi_{SName,SA}(\sigma_{CName='清华CC'} \bowtie CP \bowtie \sigma_{PS='科技' \vee PS='小说'}(P) \bowtie SP \bowtie S)$

(3) $\Pi_{SName}(\sigma_{SA='北京'}(S) \bowtie \sigma_{CName='人民'}(C) \bowtie CP \bowtie \sigma_{PS='小说'}(P) \bowtie SP)$

(4) $\Pi_{SName}(S \bowtie (\Pi_{PNO}(SP) - \Pi_{PNO}(\sigma_{CName='北大'}(C) \bowtie CP \bowtie \sigma_{PS='科技'}(P))))$

(5) Π_{SName} [redacted] $(S \bowtie (SP \div \Pi_{PNO}(\sigma_{CName='清华'}(C) \bowtie CP \bowtie \sigma_{PS='科技'}(P))))$

2.(1)

S

SNO	SName	SA
S1	P.A	北京

SP

SNO	PNO	DATE
S1	P1	
S1	P2	

P

PNO	PName	PS
P1		科技
P2		小说

(2)

S

SNO	SName	SA
¬ S1	P.A	

SP

SNO	PNO	DATE
S1	P1	

P

PNO	PName	PS
P1		小说

CP

CNO	PNO	PD
C1	P1	

C

CNO	CName	CA
C1	人民	

(3)

S

SNO	CName	SA
S1	P.A	P.北京

SP

SNO	PNO	DATE
S1	P1	
S1	P2	

CP

CNO	PNO	PD
C1	P1	
C1	P2	

C

CNO	CName	CA
C1	商务	

P

PNO	PName	PS
P1		科技
P2		科技

三.(1)

```
select S.SName, count(SP.PNO)
from S, SP
where S.SNO = SP.SNO
group by SP.SNO
having count(SP.PNO) >= 10;
```

(2)

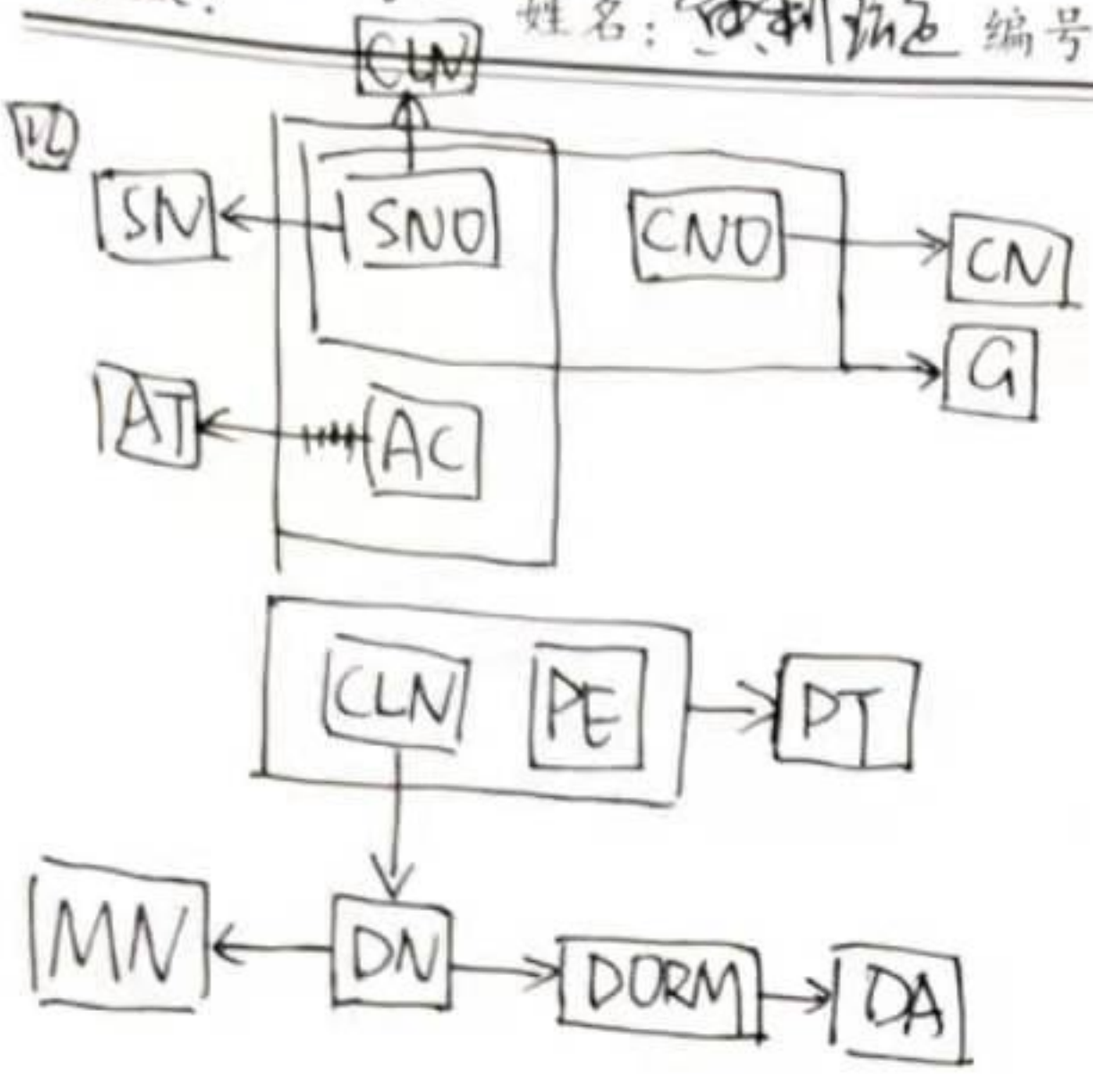
```
select SName
from S, SP, P
where S.SNO = SP.SNO and PName = 'PG' and P.PNO = SP.PNO
  and DATE <= ALL (select DATE
                   from SP, P
                   where PName = 'PG' and P.PNO = SP.PNO);
```

③
```
select SName, SA
from S, C, CP, SP, P
where CName='清华' and C.CNO=CP.CNO and CP.PNO=SP.PNO
  and S.SNO=SP.SNO and SP.PNO=P.PNO and PS='科技';
```

④
```
select SName
from S, SP
where S.SNO=SP.SNO and SP.SNO in
    (select SNO
     from P, SP
     where P.PNO=SP.PNO and PS='科技' and SNO in
    (select SNO
     from P, SP
     where P.PNO=SP.PNO and PS='小说'));
```

⑤
```
select SName, SA
from SP First, S
where S.SNO = First.SNO and not exists
    (select *
     from SP Second, C, CP
     where CName='高数' and C.CNO=CP.CNO and CP.PNO=Second.PNO
    and not exists
     (select *
      from SP Third
      where Third.PNO=Second.PNO and Third.SNO=First.SNO));
```

四



都满足3NF

分解后：

学生关系：S<SNO, SN, CLN>

GR<SNO, CNO, G>

C<CNO, CN>

A<SNO, AC, AT>

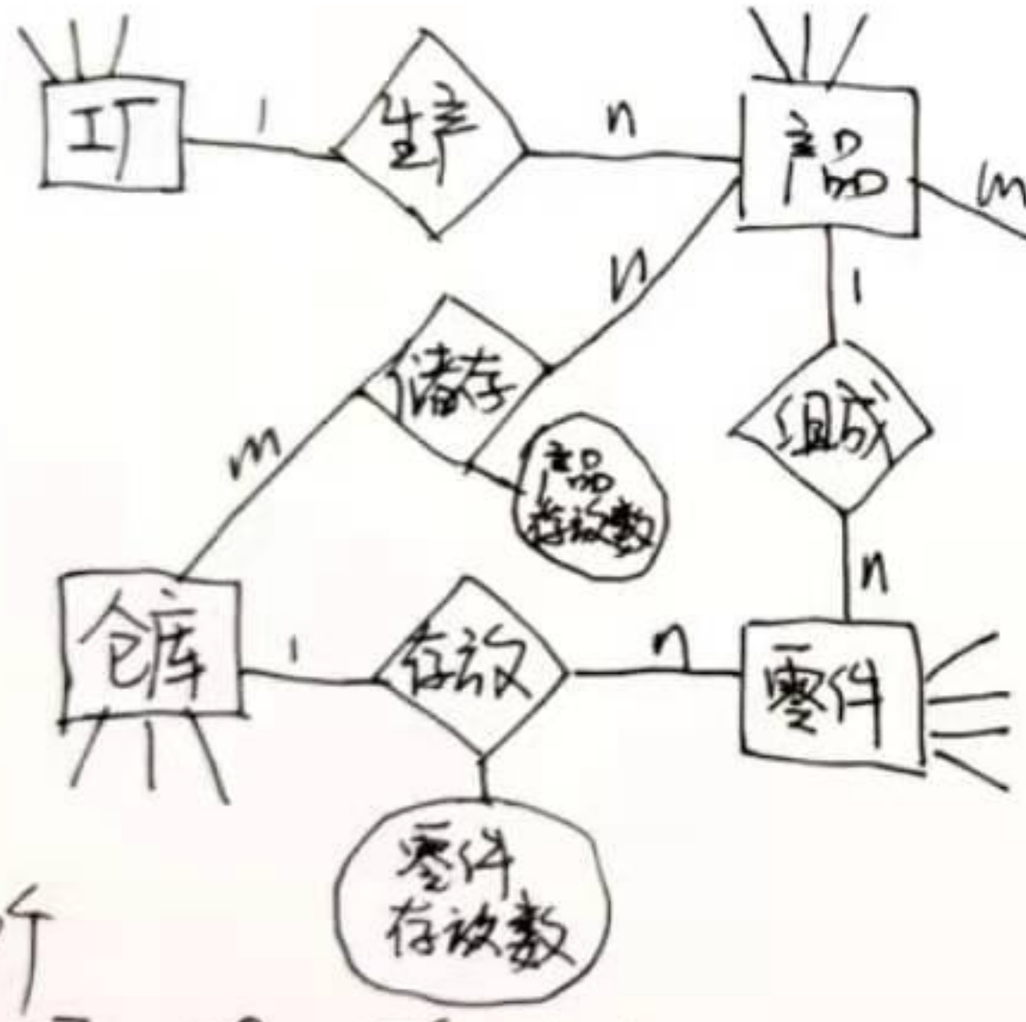
班级关系：CL<CLN, DN>

D<DN, MN, DORM>

DO<DORM, DA>

P<CLN, PE, PT>

五.



共10个

工厂<工厂号, 工厂名, 地址>

产品<产品号, 产品名, 产品数量>

零件<零件号, 零件名, 零件数量, 颜色>

仓库<仓库号, 仓库名, 地址>

经销商<经销商号, 姓名, 所在城市, 联系电话>

生产<产品号, 工厂号> 组成<零件号, 产品号> 销售<产品号, 经销商号, 销售量>

储存<产品号, 仓库号, 产品存放量> 存放<零件号, 仓库号, 零件存放量>

满足3NF至少需要7个

工厂<工厂号, 工厂名, 地址>

产品<产品号, 产品名, 产品数量, 工厂号>

零件<零件号, 零件名, 零件数量, 颜色, 产品号, 仓库号, 零件存放量>

仓库< >

经销商< >

储存< >

销售< >